Section A: Data Fundamentals

(1) - Data base is a system used to store, ~~data, and~~ manage, & safe keep data.
Example :- e health system in hospital
→ Bank - financial system
Reason why it is important :- data base
help ~~to~~ ~~Retrive~~ Retrival of data easily
→ data is no lost easily
~~→ can be stored for a long time~~
→ Data base is a key to an operation.

(2) Data were house :- is a platform that keep & store different / multiple data from different sources. for a long time.
→ Data can be stored for a long time
→ Help Retrive ~~old~~ old data / historical data

Companies use data were house ~~because~~ for fast and efficient data reporting
→ help in future / decision making for the company.
- nosty structural data

(3) Data Lake - help to store large volume data in its original format at any scale
→ can also store unstructured data

* when ~~all~~ usual business prefer data lock over data were house
→ when large file need to be stored for example hospital file archiving
→ It is cost effective.

(b) Data Mart → is drawn from a data warehouse (or subset) focusing on a single / particular subject area. eg. departments i.e. HR, Finance

→ Data Mart → take data from warehouse and divide it into smaller controllable sections

can be used by business analysts / managers

(5) (a) - Structural Data → data that is organised into tables - rows & columns. can be easily analysed

e.g. Can be in rows & columns. of Name & adresses of Employees.

e.g. spreadsheet, application forms, Hospital records

(b) Semi-structural → data is not strictly in format, tag or labels but flexible.

(c) Un structural data → has no specific format or organisation. → hard to store

eg. Voice recordings, social media posts, media files, PDF document

(6) Data modelling → is a process / way to create data prescriptions & analysing different data.

→ It improve data quality.

→ help make decisions in a company.

Data model → Relational model - tables

→ Hierarchical model →

(7) → Data mining → ~~To use~~ the use of Machines to ~~algorithms~~ analyze ~~analyze~~ & uncover patterns & other values. From larger data set.

→ to understand Complex data

→ help in predicting models

* Retail - use data mining to analyze ~~analyze~~ historical Purchase to identify Product Purchase pattern

* financial institution use Data mining to identify fraud. Sept suspicious

(8) Database management systems → is the use of software system to ~~manag~~ manage the information in the data base. [systems used to access the data base.]

(a) MySQL → adopted in open source in Relational DBMS. such as web application.

(b) POSTgreSQL → object Relational database systems, mostly Applied in Geographical ~~data~~ data

(c) Oracle Database - for larger ~~scale~~ scale enterprise industries like ~~financial~~ financial institution & healthcare

(d) MongoDB → stores flexible data, with ~~a~~ Rapid Changing models

(9) ETL → process of transferring raw ~~data~~ data in to valuable ~~info~~ insight

(a) Extraction → connect data from different sources. including database.

(b) Transform → include steps such as - Cleaning data normalizing - into ~~state~~ standard formats & Enriching (adding information) / Aggregation,

(d) Loading → load in to a data storage to warehouse / data lake → to be use in reports.

→ ~~It is also~~ the step is important / ETLU important because it convert data from multiple source into usable format.

(10) Data storage format are used to encode data into a file

(a) CSV → (Comma-separated values) → a text based tabular data format.

→ Each line represent a row data & field of of rows are separated by commas.

→ use format for exporting spreadsheet or data base.

(b) JSON → (JavaScript Object Notation) — a light weight text-based format for storing & sharing data, human & computer can read & support nor complex data structure than CSV.

(c) Parquet → A columnar data file

Useful for larger data sets & real time big data processing

→ more efficient than rows as due to advance technology conversion technology minimizes storage requirements.

→ for big data warehousing, querying & Petrol

→ binary data file format (not text)

* human can't open & read parquet.

(D) AVRO - data serialization system, binary row based format - used in big data environment for data storage

Section B

11. ~~AI → use of machines~~ Making ~~of~~ computers

11. AI → Use of machines to simulate human intelligence, to solving problems & ~~[illegible]~~ tasks

(a) Narrow AI → designed to single task, work within their scope. eg. Apple Siri, Chat GPT.

(b) General AI → have ~~ability~~ ability to understand, learn & apply knowledge in different tasks.

(12) - Machine Learning (ML) - is a subset of AI, that focuses on algorithms that analyse & learn the patterns of training data.

→ intended this performance as they are exposed to more data

→ traditional programming, a developer write ~~[illegible]~~ code / instruction for computer to follow.

∴ ML as ~~[illegible]~~ ~~not~~ machine ~~[illegible]~~ learning - systems learning those rule ~~[illegible]~~ automatically from data.

3 main types of ML (~~[illegible]~~ machine learning).

① → supervised → model is trained using labeled data → system learns the relationship between input & correct outputs.

eg: Email spam detection

② → unsupervised → the model work ~~[illegible]~~ with unlabeled data → ~~[illegible]~~ discover ~~hidden~~ hidden data relationships

③ Reinforcement → system ~~[illegible]~~ learns by interacting with the environment & receiving rewards/penalties → making decisions to learn from feedback

(13) Deep learning is a system under Machine learning. ~~that~~ uses ~~multi~~ multi

13 Deep learning is a system that uses multi-layer Algorithm, & uses large data sets.

* Machine learning & deep learning differs from how each algorithm learns & the amount of data they use.
* AI - focuses on mimicing human intelligence
* ML (machine learning) - learn from data not specifically programed.
* Deep learning - use deep ~~neuro~~ ~~neural~~ Neural networks ~~later~~ to learn complex data

Neural networks → subset of machine learning
at is a backbone of deep learning.
→ they are ~~neural~~ Neural because they mimic the ~~brain~~ signal of neurons of the brain.
~~Neuron~~ Neural - has input layer - receives data
→ hidden layer - process data
output layer - produce results
→ data is fed in to the network, weights each connection between neurons, data ~~moves~~ moves layer by layer till output

(14) Natural Language Processing (NLP) - enable computer & bridge a gap ~~for~~ for computer to understand, interpret & generate human language
→ it detect emotions, general human response
Translate between ~~by~~ languages
eg - voice Assistance - Google Assistance
→ Auto correct - check spelling & grammar
→ Chat bot / ↔ provide answers

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(15). Large language model (LLM). -

→ ~~dest~~ designed to ~~[illegible]~~ understand & generate, ~~too~~ & work with human language

→ LLM's → ~~trained~~ are trained on large data sets ~~[illegible]~~ (large scale size of data)

→ Traditional LM uses simple Algorithm. Traditional LM is limited to understanding context. However ~~[illegible]~~ LLM can understand context, tone & relationship between sentences.

Two LLMs

(1) → GPT-4 (Open AI). - use to Copilot Reasoning. (e.g. Chat GPT) ~~[illegible]~~.

(2) → Claude 3 (Anthropic) used for Content Creation, Coding & Insight Analysis of large documents.

(16). - Generative AI - type of AI that Create new Content, such as ~~text~~ text, code or image, learning patterns from massive existing data.

→ Synthesizing patterns from large / massive data it does not memorise by heart. the Probability of what ~~[illegible]~~ Comes next

→ General output:

3 popular AI TOOLS → (1) Chat GPT - can answer questions with essays & summarise content.

(2) DALL-E 3 (Open AI) → Create image, used in design workflows & Creative Project

(3) Github copilot → ~~[illegible]~~ in Codes editors. Provide Real time Content.

(17) Training data — information that is used to teach a machine learning model how to make tasks & recognise patterns.

→ Quality of training data — can determine how well the AI model performs

What could go wrong if training is biased/incorrect

→ Incorrect predictions
→ model ~~[illegible]~~ have favoritism to one group and lead to poor outcome
→ fails in real world

(18). Computer vision → the AI that enables computers to see, understand & interpret visual information from the world, replicating human sight.

- they process image as numerical data & AI cameras capture image into digital form (pixels)
- feature extraction - use algorithm & models like Convolutional Neural networks, that detect shapes, textures & colour

Real world → (1) facial recognition for security

(2) Autonomous vehicle → using camera car self driving & detect marking or objects.

(3) Healthcare Imaging → X-Ray, MRI

Section C:

(19) API - Application programming ~~Interface~~ Interface is a set of rules that allow different software applications to ~~exchange~~ exchange & communicate data with each other.

Codes fully API → client App make request for a specific object

→ ~~API~~ Server - prepared

→ API documentation tells what data/information is available

→ API → gives order

→ APIs allow different applications (languages) to work together seamlessly

→ Save time, improve scalability & flexibility.

(20) API key - is a unique code that identifies & authenticates the application when it makes request to an API.

Purpose: ~~use~~ for identification, for Authentication & ~~usage tracking~~ usage tracking

→ API key is treated like password.

→ If exposed publicly.

→ data loss - server can delete

→ service disruption.

REST API → is a type of API that follows a specific architecture style to building web services,

- Allow system to communicate Internet ~~by~~ using standard HTTP protocol,

Representational

REST → Re ~~Representational~~ State Transfer
- set principles to guide how web should be ~~designed~~ designed. to be simple, scalable & stateless

REST APIs use HTTP methods to perform actions on Resources. (Patients / Results)
main HTTP methods

① GET - Retrieve ~~data~~ data.
used to Fetch information. eg: patient list

② POST - Create new data. - Add new record or patient list.

③ PUT → update Existing data.
→ used to update & replace an existing resource. eg update patients information eg. (address) Address

④ Delete - Remove data

(22) → JSON - Java-script-Object Notation. - is a light weight,
Why JSON is Commonly used in APIs
→ simplicity & Reading - it is easy to human Read.
→ Lightweight Nature
→ Native Javascript support.
→ universal Compatibility

JSON - {"name": "John",
"age": 30
"Email": john@gmail.com
"is Student": True.

(23) Prompt Engineering - is the art & science of testing & refining input instructions (Prompt) to guide AI ~~model~~ Model.

→ AI work Respond to "Prompt" – Instructor

Why is it a important skill / to get :

→ Precision & Quality Result
→ Efficiency – produce usable content & code
→ Context Control :
  Eg help to improve prompt.

Example : write something about Hypertension & Diabetes
It is wrong
Improved prompt : Explain the difference between hypertension & Diabetes.

(2nd) the Map function is a function that applies a specific transformation function to every element within the list & Return a new Collection Containing Results.
- used to clean data
- Replace verbose loops with single line of code

Example : python :
```
# Function to Apply
def - Square (n):
    Return n*n
numbers = [1,2,3,4]
# Applying map (:)
Squared_numbers = map (Square, numbers).
# Converting Results list
Result = list (Squared_numbers).
Print (Result) # output [1,4,9,16]
```

(25) Cloud Computing - is the on demand delivery of IT Resources (server, storage, database, networking, software). over the internet with pay as you use pricing

→ 3 main cloud service models

→ Iaas → Infrastructure as a service → provide fundamental computing resources, virtual servers, storage & networking.
eg Amazon EC2.

→ Paas → Platform as a Service → delivers a cloud environment that includes hardware & software tools. allowing developer to build, test & deploy applications without managing underlying infrastructure eg Google App Engine.

→ Saas - Software as a service → Delivery fully managed, ready-to-use software applications over the internet, often accessible via web browser with provider handling all maintenance eg Microsoft office 365.

(26) VCS → Version control system → Software tool that tracks & manages any change made to a codebase/project file over time.

Git → an open source, distributed version control system. Every user has a full copy of the project history on their computer.
→ for collaboration - Multiple developers can work on a project at the same time.
→ history tracking - Every change is recorded & tracked
→ backup & safety
eg Github.

Save snapshot of your Project.

Commit → Save point

Branch → is a separate version of your project where you can work independently

(27) Jupyter Notebook → an interactive web based tool that allows you to write & run code, display results, & explanations (text, images, charts) all in one document.

→ Jupyter notebooks are most commonly used with python.

- Why popular → Interactive Coding
  - → Combine
  - Data visualization - create charts & graphs using libraries like seaborn & pandas
  - support Data analysis tools - numpy

(28) What is python → high level programming language known for its simple English-like syntax

Python ~~often~~ often used for web development & video game development

- It is easy to learn & use: clean syntax.
- → Extensive libraries & frameworks.
- → versatile - can be used for data analysis

Common python libraries in data work.

→ Numpy - Numerical python → provide support for large arrays

→ Pandas → introduces data structures like Dataframe which allow users to clean, transform & analyze structured data easily.

→ Matplotlib → Data visualization - create graphs, charts & plots.

→ Scikit learn - provide simple & efficient tool for data mining & data analysis

SECTION D

(29) Data privacy → is the right the control how information is collected, used & shared, ensuring security &
* It is implemented to prevent theft/& harm
* protect personal information

→ GDPR → General Data Protection Regulation.
→ major data law introduced in European Union (EU) ~~in 2016~~ adopted in 2016 & came to effect in 2018.
It applies to countries in European Economic Area (EEA). and all countries ~~that~~ ~~trade~~ that offer services to EU/EEA ~~countries~~.

Key ~~Privacy~~ Principle → transparency
→ Data minimization
→ Strict consent - organizations must be clear, get permission before using data
→ Accountability →

(3) Data Ethics → Analyze, evaluate & ~~establish~~ establishing moral obligations of how data is collected, & stored & used.
- It ~~ensures~~ ensures data practices that respect human rights, ~~privacy~~ privacy & fairness

Issues of machine Data/AI

(1) Biased training process → may intensify ~~focus~~ tension certain groups over others, due to it training on biased historical data → people may be denied jobs based on race, gender / background)
(2) Unauthorized Data Sharing → DIT violate personal autonomy & privacy rights, causing surveillance issues.

(3) Exploitation of user data for political manipulation
* ~~Gen~~ AI used to generate fake news, misleading content.
* It can change perception & create confusion

(31) Data governance → is a framework of policies, processes, roles & standards that ensure an organization's data is secure, high quality & accessible throughout its lifecycle

Key components of good data governance framework

→ Data strategy & policies – define rules governing data collection, usage, storage & retention
→ Roles & accountability → clear assignment of responsibility
→ Data quality management – ensure accuracy / reliability of data
→ Data security & privacy – protect sensitive data
→ Data ownership →

(32) Bias in AI → Refers to unfair errors in AI system output. Caused by prejudiced assumptions in its training data, design or development
- Data bias → training data is non-representative, incomplete or contains historical prejudice
- Algorithm bias → algorithm favors certain outcome.
- Stereotyping bias →

Example → A hiring tool – more AI was trained that male are preferred for ~~certain~~ women. System ~~do~~ ~~select~~ selected ~~any~~ males. women ~~are~~ were rejected based on age
→ Discrimination – unfairness
→ Legal & ethical problem

Role

(33) (i) Data Analyst → look at past data — what happens
(ii) Data Science - Predict the future outcome → what will happen
(iii) Data Engineer → Build systems & Infrastructure
(iv) AI Engineer Applies AI in real product.

	Data analyst	Data scientist	Data Engineer	AI Engineer
Role	→ Cleaning data, Creating Report, Dashboard, Identify Trends	Data modeling, Predictions, Advanced Analysis	Create data pipelines, manages databases	Build AI models, optimize its performance
Skills	Data visualization, Excel, SQL	Advance SQL	Infrastructure, Cloud Platform	ML/DL, Deploy models using API
Tools	SQL, Excel, Power BI, Python	Python, R, Scikit-learn, TensorFlow	Apache, Kafka, Airflow, AWS	TensorFlow, PyTorch, Hugging Face, Docker

(34) BI - Business Intelligence → Refer to the procedural and technical methods, which involve data mining, analysis and visualization → used to convert raw data into actionable insights.

~~Power BI~~

(1) Power BI → tool that connects multiple data sources. (Excel, databases, cloud services) creates interactive ~~Cloud~~ Reports.
A → help managers track KPIs, to make informed decisions.

(2) Tableau → enable user to build charts, graphs & dashboards with drag & drop features;
→ help business to explore data visually & communicate insights clearly.

Section E

(35) General Data protection Regulation → It is referred as the ~~Private~~ Strict law/regulation.
→ Just understand on what ~~Prot~~ Protected Personal data Lecture 2018.

(36) In a new in data Analysis → 1 Last worst ~~question~~ question ~~Study~~ a bit mind blowing that After data base there is a overhead mind ~~finding~~ thinking of data we ~~have~~ have CS data lecture, Data mint

(37) I think that data is everywhere, Everything the next outcomes are visualised somewhere all decisions are made.
- In Academics there a AI tools introduced all it is changing Institution policies on how to deal with AI.
- Prompt Ask AI about their health and Care or not related.

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